PID MOTOR CONTROL LOOP

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github.com/gallicchio/motor-pid
circuit, scripts, schematic, this board

LAB 10L: the fader's wiper is held where TARGET says, by P, D and I.

POWER: +15, GND, −15 V, 1 A each. Both green lamps light.
FADER: motor to MOTOR + GND; track ends to POT + and −, wiper to 2. POSITION must read +7.7 V at the + end. If the loop runs away, swap the two motor wires.
SWITCHES to start: TARGET SOURCE 1 (knob), LOOP 1 (open), D OFF 1, I OFF 1, D CAPACITOR 1 (22 nF). Trimmers fully anticlockwise.
THEN: LOOP 3 closes it. Raise P GAIN until it rings (~12 Hz), switch D IN and turn D GAIN up; back to low P, I IN, I GAIN up.
SQUARE WAVE: TARGET SOURCE 3, feed TARGET IN +/−0.5 V at 0.2 Hz.
STALL: the 0.35 A fuse opens in seconds, red lamps light; resets.
PROBE LOOPS on every node, GND x4.

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